A. Funky Numbers

time limit per test 2 seconds
memory limit per test 256 megabytes
input stdin
output stdout

Rating: 1300 | Tags: binary search, brute force, implementation | https://codeforces.com/problemset/problem/192/A

As you very well know, this year's funkiest numbers are so called triangular numbers (that is, integers that are representable as , where *k* is some positive integer), and the coolest numbers are those that are representable as a sum of two triangular numbers.

$$\frac{k(k+1)}{2}$$

A well-known hipster Andrew adores everything funky and cool but unfortunately, he isn't good at maths. Given number *n*, help him define whether this number can be represented by a sum of two triangular numbers (not necessarily different)!

Input

The first input line contains an integer *n* ($1 \le n \le 10^9$).

Output

Print "YES" (without the quotes), if *n* can be represented as a sum of two triangular numbers, otherwise print "NO" (without the quotes).

Examples

Input
256

Output
YES

Input
512

Output
NO

Note

In the first sample number .

$$256 = \frac{2 \cdot 3}{2} + \frac{22 \cdot 23}{2}$$

In the second sample number 512 can not be represented as a sum of two triangular numbers.